

Informed Trading in the Index Option Market

Andreas Kaeck[§] Vincent van Kervel* Norman J. Seeger[†]

[§]University of Sussex Business School, United Kingdom, e-mail: ak486@sussex.ac.uk

*Escuela de Administración, Pontificia Universidad Católica de Chile, e-mail: vincentvankervel@gmail.com

[†]VU Amsterdam, The Netherlands, e-mail: n.j.seeger@vu.nl

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Abstract. We propose a structural vector autoregressive (VAR) model of informed trading in option markets to analyze whether investors use options to trade on private information about the underlying price and/or the underlying's volatility. We decompose option order flow into exposures to the underlying asset (through the option delta) and its volatility (through the option vega). Our proposed methodological framework facilitates meaningful aggregation of option order flows for different strike prices and maturities, and increases statistical power to identify informed trading. A fitted model confirms that S&P500 option trades are indeed informative about changes in both the underlying and volatility.

Key words: Options, informed trading, price impact

1. Introduction

It is well known that investors may use option markets to exploit private information on future movements of both the underlying asset and its volatility.¹ Showing empirically the magnitude of both forces is, however, challenging. Informed trading on a single underlying is spread across a large cross section of sometimes hundreds of options which are all traded simultaneously, but differ in strike price, expiration date, and option type (put or call). To make statements about informed trading, one needs to meaningfully aggregate results across options and account for the high correlations in option returns and order flows. Another challenge arises because volatility itself is not directly observable and must be estimated from either option prices or time series data. The main goal of this paper is to address these challenges in a parsimonious model that jointly estimates informed option trading on both the direction and the volatility of the underlying asset for the entire cross section of options.

The economic motivation for our analysis is based on models of strategic informed trading (e.g., Kyle (1985), Glosten and Milgrom (1985)). Suppose that each period there is an innovation in the fundamental value of the underlying asset. This innovation is unobservable to the public, but some investors may receive a noisy signal of the realization or its variance. These investors are informed and can profitably trade options to exploit their information. If the market is rational, and aware of the presence of informed traders, it can use observable order flows to infer the realizations of these signals and update prices accordingly. For example, if informed investors believe future volatility will increase, they trade to increase their volatility exposure by buying both put and call contracts.² The market, upon observing this buying pressure, increases its expectation of the fundamental volatility and uses the updated belief to increase prices of all puts and calls. If informed traders believe the price of the underlying will increase, they increase exposure to the underlying by buying calls and selling puts. In this case, the market observes order flow generating a positive exposure to the underlying (but little to the volatility), and will

¹At the market level, shocks to volatility (or uncertainty shocks) have large consequences for real economic activity, including future productivity growth, industrial production, and employment (see e.g., Bloom (2009) and Berger, Dew-Becker, and Giglio (2019)).

²By buying an at-the-money straddle using a long put and a long call, the trade exposure is mainly to volatility as the exposure to the underlying cancels out.

update its expectation of the underlying, and of option prices, accordingly.

Analyzing order flows of all option contracts in a joint framework is crucial for several reasons. First, cross-option strategies are extremely common. [Fahlenbrach and Sandås \(2010\)](#) find that cross-option trades account for more than 75% of volume in their sample of FTSE 100 options. Second, a trade in a single option gives exposure to both the underlying and the volatility component. We can only infer the net direction of the aggregate option market (long or short) by pooling exposures to both components across options. And third, theoretical models show that informed traders act strategically and typically trade in several correlated assets simultaneously, see e.g., [Biais and Hillion \(1994\)](#) and [Boulatov, Hendershott, and Livdan \(2012\)](#). This implies that informed traders may spread their trades across many option contracts and only a joint analysis of order flows can measure their overall activity.

The main contribution of this paper is a methodological framework that recognizes that any option trade provides exposure to the underlying asset (through the option delta) and to its volatility (through the option vega). Accordingly, we disentangle the effect of an option trade on an underlying's price or volatility by constructing two order flow exposures which multiply option net order flow (defined as buyer-originated minus seller-originated volume) by the option delta or vega, which we coin "delta order flow" and "vega order flow". The advantage is that both order flows are measured in a unit that can be meaningfully aggregated over all trades in the cross section of options. We then relate the two option order flows to changes in the underlying price and its volatility in a vector autoregressive (VAR) model. We further add a fifth equation with the net order flow in the underlying asset, which allows for a direct comparison of informed trading in the underlying asset and the aggregate option market. Our framework extends the seminal work of [Hasbrouck \(1991\)](#) who estimates a two-equation VAR model relating order flows in an asset to its price changes. We address the challenge that volatility is unobservable by linking a structural option pricing model to the model-free VIX framework. This allows us to link the unobservable volatility to an observable volatility index, which yields consistent delta and vega estimates and neatly integrates into the VAR system.

Our framework offers three key advantages for analyzing informed trading in options markets.

First, we estimate a single VAR model that uses all trades in the cross section of options. Our approach correctly deals with cross-option strategies, and is representative of informed trading and liquidity of the aggregate option market. We effectively apply a data reduction approach by imposing economic structure from a theoretical option pricing model to summarize the information content in a large number of order flows in just two variables – delta and vega order flow. This structure allows us to disentangle volatility and price movements, and to increase the statistical power to detect informed trading on both components in a unified framework. Second, the model yields a novel price impact parameter of vega order flow on the level of volatility which represents the (il)liquidity of volatility speculation. In equilibrium, market makers observe the vega order flow, and respond by adjusting option prices (and implied volatilities). Our estimated price impact of vega order flow on volatility captures this adjustment, and can be interpreted as a measure of the degree of informed volatility trading. And third, the model accounts for the strong return and order flow correlations across options. Specifically, a trade in one option affects returns of other options through its impact on the underlying price and volatility. By aggregating the *net* delta and vega order flows across options we capture cross-option trading strategies.

Our empirical analysis employs the five-equation VAR model using trade and quote data for SPX options written on the S&P 500 index. Our order flow for the underlying asset is constructed from the ETF on the S&P500 (SPY) and we use the VIX index (which we construct at high frequency) as the volatility index. We construct the delta and vega order flow variables at an hourly frequency and estimate the model using all trading days in 2014.

The main empirical findings are as follows. In a given hour, a one standard deviation shock to vega order flow increases the volatility by 0.07 percentage points (e.g., increasing the average volatility of 11.50% to 11.57%). This effect is 14% of the hourly standard deviation of volatility changes, and reflects private information in the volatility component of option order flow. Together, the delta and vega order flow explain 25% of the variation in volatility, leaving 75% to public information arrival. Similarly, a one standard deviation delta order flow shock increases the underlying by \$0.98 (or 5.1 basis points relative to an average price of \$1,930). The effect is larger for a one standard deviation shock in the ETF order flow, which moves the underlying

by \$1,71. In fact, the *per dollar* price impact of ETF order flow is 3.4 times larger than the option order flow, after adjusting for the fact that delta order flow shocks have a larger standard deviation than ETF shocks. The higher price impact of ETF flow indicates a higher degree of informed trading in this market, which seems surprising as the SPY ETF is very liquid compared to the SPX options. A potential explanation is that the size of option trades is generally an order of magnitude larger than that of ETF trades, which suggests that the ETF attracts relatively small but informed traders whereas option markets attract investors wanting to trade in size.

Other results of our VAR model estimation align with the existing literature. The order flow in the ETF is highly correlated with the delta order flow, which suggests investors use both options and the ETF when looking for directional exposure. All three order flow variables are highly autocorrelated at the hourly frequency. The model also confirms the well known leverage effect, which is the negative correlation between changes in the underlying price and volatility. This effect emphasizes the importance of modeling both processes jointly, which otherwise would cause biases in parameter estimates.

Our model imposes a strong economic structure on the data through three crucial assumptions. We validate these assumptions with several tests. First, using a Wald test, we test whether the delta and vega price impacts vary across options with different expiration dates or strike prices. We do not reject equality for vega flows, meaning the vega component of order flows can indeed be pooled across options. The delta flow does differ significantly across groups of options, and in particular we find that options with a short time to maturity and low moneyness have a higher price impact. Second, we test the information loss of our model, which uses a two-factor option pricing model (underlying and volatility) to summarize the price changes across all options with different strike prices and maturities. The analysis, following Bakshi, Cao, and Chen (2000), indicates that our model explains 99.2% of the variation in individual option price changes. Together, the two tests seem to validate our imposed structure, and provide statistical support for a five-equation system to capture the information content of hundreds of option prices and order flows. And third, we test the assumption of the model stemming from the ordering of the equations in VAR models. In the robustness section we consider several different orderings of

the equations, and find that all results hold in alternative model specifications.

Compared to the existing literature on option order flow (which we summarize below) our main results are based on hourly rather than daily or weekly observation frequencies, and therefore provide new insights into the impact of intra-daily option order flow on option prices. From a methodological point of view, our framework can be extended to higher frequencies which we investigate in a robustness section. Studying order flows at one minute intervals, we find that the high bid-ask spreads and tick sizes in option prices provide an empirical challenge to studying option order flows at very high frequencies. In essence, the delta-vega approximation of option price changes are empirically much less successful since the two factor approximation explains less than 50% of option price changes at the one-minute level.

Our paper is most closely related to [Pan and Poteshman \(2006\)](#), who show that option order flow is informative about the directionality of the underlying, and [Ni, Pan, and Poteshman \(2008\)](#), who focus on the impact of a variable similar to vega order flow on future realized volatility. Our work complements these papers, and offers several methodological contributions and novel empirical findings. First, and most importantly, our framework jointly models the informational content of both option order flow components.³ Second, [Ni, Pan, and Poteshman \(2008\)](#) focus on an at-the-money option, whereas our VAR model aggregates volatility information from the whole cross section of option prices. The imposed structure yields additional statistical power. Third, our empirical findings are based on order flows that are constructed from publicly available trade and quote data, whereas [Pan and Poteshman \(2006\)](#) and [Ni, Pan, and Poteshman \(2008\)](#) use non-public data of non-market maker position-opening order flow disaggregated by investor type. Fourth, our methodology measures permanent price changes due to informed trading, and is not affected by temporary demand pressures. And fifth, the focal point of our empirical study is informed trading on market-wide volatility rather than the volatility of individual stocks.

The aim of our paper is also closely related to [Chordia, Kurov, Muravyev, and Subrahmanyam \(2019\)](#) who study the impact of weekly index option order flow on subsequent S&P 500 index returns. Their study differs from ours in both methodology and empirical findings. [Chordia,](#)

³This approach provides new insights regarding the relative information content of option and spot markets by comparing the price impact of various order flow variables (as in [Easley, O'Hara, and Srinivas \(1998\)](#)).

Kurov, Muravyev, and Subrahmanyam (2019) find that SPX order flow is uninformative about future S&P 500 index returns, whereas the order flow of other index options, namely options on ISE 250 Index, Russell 2000 Index, Nasdaq-100 Index and the S&P Mid Cap Index 400 (all traded on the International Securities Exchange (ISE)), do provide significant predictive power. In contrast, our results offer strong support for a direct connection between order flow in SPX and S&P 500 index returns and their volatility. In addition, Chordia, Kurov, Muravyev, and Subrahmanyam (2019) find that high net put order flow of ISE index options predicts negative index returns, and they attribute this to informed market makers. We find strong support for the opposite effect which is in line with Chen, Joslin, and Ni (2018), who show that monthly net order flow in deep-out-of-the-money SPX put options predicts low index returns. We generalize this result as our order flow variables are constructed from all option trades. In addition, Chen, Joslin, and Ni (2018) and Chordia, Kurov, Muravyev, and Subrahmanyam (2019) focus on position opening trades, but this information is not public. We sign option trades and weigh them by their price and volatility exposure, which is directly observable by all market participants.

The literature on analyzing the impact of option trades on the volatility of the underlying is relatively scarce. Bollen and Whaley (2004) show that option-implied volatilities are positively related to the net buying pressure for both S&P 500 index option and options on individual stocks (see also Garleanu, Pedersen, and Poteshman (2009)).⁴ Puhan (2014) combines daily changes in open interest of different option categories to identify straddle positions (which provide exposure to volatility). She relates open interest changes to future realized volatility and finds significant effects, especially around macroeconomic news announcements. Figlewski and Frommherz (2017) extract the volatility processes for groups of options sorted on moneyness and put-call characteristics from high frequency option quote data on the S&P500, the E-mini and the SPDR ETF. They show that out-of-the-money options lead innovations in volatility.

A substantial literature focuses on the contribution of option markets to price discovery. In extensions of the Hasbrouck (1995) information shares analysis to option markets, several papers find modest price discovery through option quotes (e.g., Chakravarty, Gulen, and Mayhew

⁴They also find that approximately 20% of the price impact is reversed the next day, suggesting that a transitory price pressure is an important component of the positive price impact.

(2004), Holowczak, Simaan, and Wu (2006) and Rourke (2013)). More recently, Patel, Putniņš, Michayluk, and Foley (2016) propose an improved version of the Hasbrouck (1995) information share measure that accounts better for the typical noise in option quotes. They find that options contribute 33% to price discovery. Using a rather different approach, Muravyev, Pearson, and Broussard (2013) find that violations of put-call parity are resolved by adjustments of option prices, which they interpret as evidence of no price discovery in the option markets.

2. A Price Impact Model for Option Markets

This section presents a novel methodology to measure the price impact of option order flows. We extend the VAR model of Hasbrouck (1991) to option markets and explicitly take into account (i) the cross-option correlations in order flows, (ii) the cross-option price impacts, and (iii) the fact that options can be used to speculate on both the underlying as well as its volatility.

Consider an asset on which N different option contracts are traded. Individual options are indexed by $n = 1, \dots, N$ according to the characteristics strike price, expiration date and put/call identifier. We recognize that the main determinants of option price changes are the changes in the value of the underlying and its volatility (e.g., Bakshi, Cao, and Chen (1997)).⁵ The option price change can therefore be written as

$$d_{n,t}^o = \Delta_{n,t-1} d_t^u + \nu_{n,t-1} d_t^v + \epsilon_{n,t}, \quad (1)$$

where $d_{n,t+1}^o = P_{n,t+1} - P_{n,t}$ is the price change of option n at time $t + 1$, $\Delta_{n,t}$ and $\nu_{n,t}$ are the delta and vega of the option, and d^u and d^v denote changes in the price of the underlying and its volatility, i.e., $d_t^u = P_t^u - P_{t-1}^u$ and $d_t^v = v_t - v_{t-1}$.⁶ The error $\epsilon_{n,t}$ contains terms of order dt of the option pricing model. Throughout the paper, we define vega as the first partial derivative of the option price with respect to volatility. One of the key challenges is that volatility is not directly observable, and therefore needs to be estimated from market data. We provide further details on

⁵In the Heston (1993) model, the impact of other input variables on option price changes are negligible at high frequencies (e.g., changes in the time to maturity or second-order price effects).

⁶Theoretically, the approximation in Equation (1) follows from Ito's Lemma.

the link between volatility and observable volatility indices in Section 3.2.

We propose to disaggregate an option trade into its exposure to the underlying asset (through the option delta) and to its volatility (through the option vega). The key advantage of this linear transformation is that option order flows are now expressed in the same units, and can be meaningfully aggregated across options. We define the time- t aggregate net dollar exposure to the underlying by x_t^Δ and to the volatility by x_t^ν :

$$x_t^\Delta = \sum_i Q_i \times \Delta_{i,t} \times BuySell_i \times P_{t-1}^u, \quad (2)$$

$$x_t^\nu = \sum_i Q_i \times \nu_{i,t} \times BuySell_i \times v_{t-1}. \quad (3)$$

We sum over all option trades (indexed by i) in the fixed time interval between $t-1$ and t . With a slight abuse of notation we denote by $\Delta_{i,t}$ and $\nu_{i,t}$ the delta and vega of the particular option trade i . Further, Q_i is the trade volume in number of options, and $BuySell_i$ a binary variable that equals 1 for a buyer originated trade and -1 for a seller originated trade. In the remainder of the paper, x_t^Δ and x_t^ν are called *delta order flow* and *vega order flow*, respectively. The delta order flow is denoted in dollars, which allows for a meaningful comparison to the order flow in the underlying asset.

In the next step, we relate the delta and vega order flows to the price changes of the underlying (d_t^u) and volatility (d_t^v) in a VAR framework. We include the dollar order flow of the SPY ETF (denoted x_t^{spy}) as an additional equation, as it is a proxy for the trading volume in the underlying asset. This approach also accounts for trading strategies that involve simultaneous trading in options and the underlying. Note that the order flow x_t^{spy} is measured in the same unit as x_t^Δ , as both represent dollar order flow exposure to the underlying. The difference, however, is that the latter is constructed from option order flows only.

The empirical model can be written as follows:

$$\begin{aligned}
d_t^u &= c_1 + \sum_{l=1} A_{1,l} d_{t-l}^u + \sum_{l=0} B_{1,l} d_{t-l}^v + \sum_{l=0} C_{1,l} x_{t-l}^{spy} + \sum_{l=0} D_{1,l} x_{t-l}^\Delta + \sum_{l=0} E_{1,l} x_{t-l}^\nu + \varepsilon_{1,t} \\
d_t^v &= c_2 + \sum_{l=1} A_{2,l} d_{t-l}^u + \sum_{l=1} B_{2,l} d_{t-l}^v + \sum_{l=0} C_{2,l} x_{t-l}^{spy} + \sum_{l=0} D_{2,l} x_{t-l}^\Delta + \sum_{l=0} E_{2,l} x_{t-l}^\nu + \varepsilon_{2,t} \\
x_t^{spy} &= c_3 + \sum_{l=1} A_{3,l} d_{t-l}^u + \sum_{l=1} B_{3,l} d_{t-l}^v + \sum_{l=1} C_{3,l} x_{t-l}^{spy} + \sum_{l=0} D_{3,l} x_{t-l}^\Delta + \sum_{l=0} E_{3,l} x_{t-l}^\nu + \varepsilon_{3,t} \\
x_t^\Delta &= c_4 + \sum_{l=1} A_{4,l} d_{t-l}^u + \sum_{l=1} B_{4,l} d_{t-l}^v + \sum_{l=1} C_{4,l} x_{t-l}^{spy} + \sum_{l=1} D_{4,l} x_{t-l}^\Delta + \sum_{l=0} E_{4,l} x_{t-l}^\nu + \varepsilon_{4,t} \\
x_t^\nu &= c_5 + \sum_{l=1} A_{5,l} d_{t-l}^u + \sum_{l=1} B_{5,l} d_{t-l}^v + \sum_{l=1} C_{5,l} x_{t-l}^{spy} + \sum_{l=1} D_{5,l} x_{t-l}^\Delta + \sum_{l=1} E_{5,l} x_{t-l}^\nu + \varepsilon_{5,t},
\end{aligned} \tag{4}$$

where $A_{i,l}, \dots, E_{i,l}$ are constant coefficients. Equation (4) emphasizes the ordering of the equations in the structural VAR. This ordering identifies the orthogonal structural innovations, as it restricts the contemporaneous innovation in one variable to affect that of another, but not the reverse.⁷ In particular, in the first equation of the system, price changes d_t^u are affected by all other variables contemporaneously but do not affect any other variables. In the last equation, the vega order flow x_t^ν contemporaneously affects all other variables but is not affected by others.

While in general the ordering is arbitrary, we believe that the model in Equation (4) is the most natural choice. First, we follow Hasbrouck (1991) by placing price changes first and order flows second. This is motivated by sequential trade models, where order flows cause returns because of asymmetric information and informed trading, whereas returns do not directly cause order flows. With regard to the ordering of the price and volatility changes d^u and d^v , we note that the former is a traded asset whereas the latter (the volatility) is not. Being a traded asset, d^u should respond fast and efficiently to new information, such that it is affected by contemporaneous innovations in volatility. The reverse does not hold, since volatility moves slower as it is only indirectly traded through a wide range of options. We also believe our model represents the most conservative ordering. Our focus is on measuring the information content in option order flows after controlling for any information captured by trading in the underlying itself. Accordingly,

⁷Our approach is equivalent to estimating the reduced form VAR, and applying a Cholesky decomposition on the residuals (see chapter 9.4 in Hasbrouck (2007)).

our proposed ordering attributes any contemporaneous information captured in for example both x_t^{spy} and x_t^Δ to the former when predicting d_t^u . While this ordering may create a bias by reducing the explanatory power of x^Δ on d^u , it is the most conservative approach for our purposes. A number of alternative specifications are studied in the robustness section.

The proposed model yields several practical advantages. First, the lags in the VAR system naturally account for any GARCH feature in the volatility process. Second, the inclusion of the volatility process allows for a simple separation of informed trading in the two components. Third, the analysis of option markets typically requires data reduction techniques as the number of actively traded options is large (considering put and call options, and different strike prices and expiration dates).⁸ To reduce the dimensionality of the data, some papers propose to model option prices as a function of time to maturity and moneyness (e.g., [Aït-Sahalia and Lo \(2000\)](#)). In this paper we apply a data reduction motivated by a two-factor option pricing model such as [Heston \(1993\)](#).

3. Empirical Setup

3.1 Data

Our trade and quote data are collected from Reuters Tick Historical. The data set consists of European-style SPX options written on the S&P 500 index and covers the time period from January 2, 2014 until December 31, 2014. Option contracts differ in strike price, put/call identifier and expiration date and are part of the third-Friday expiration cycle which includes the most liquid option contracts written on the S&P 500 index. For each option trade, we observe the exact time of the trade (to the millisecond) as well as the trade price and volume. Quote data consist of the best bid and ask prices which are used to classify each trade as a market buy or sell order.⁹ We also collect corresponding trade and quote data for the SPY ETF which we use as a tradeable

⁸Chan, Chung, and Fong (2002) also use an extension of the [Hasbrouck \(1991\)](#) model to option markets, but use only data of one put and call option.

⁹There have been some issues with signing option trades using Lee-Ready because trades can occur within the quoted prices and are often timed strategically (see [Muravyev and Pearson \(2019\)](#)). This adds measurement error to the order flow variables, but will otherwise not affect the analyses.

proxy for the underlying asset and to construct the price changes d_t^u .

We apply a range of standard filters to our data. First, we discard options with less than 7 or more than 180 calendar days to maturity. Long-term options are very infrequently traded, and short-term options are adversely affected by changes in the expiration cycle. Second, we remove individual trades for which the dollar volume exceeds \$10 million, as these are likely negotiated over the counter (OTC).¹⁰ And third, we avoid open and close rotations by restricting our analysis to trades between 8:45am to 2:45pm CT, leaving a sample of six full trading hours.

3.2 Model Design Choices and Implementation

The empirical implementation of the proposed model requires several design choices such as the sample frequency and the construction of the order flow variables x_t^Δ and x_t^ν as well as volatility changes d_t^v . We describe our approach to construct these below.

Sampling frequency: We sample data at an hourly frequency, compared to daily or weekly order flows studied in the related literature. While our data set and methodological framework allows us to conduct the empirical analysis at even higher frequencies, a careful analysis, provided in Section 5, shows that the hourly frequency provides the best compromise between conducting a high frequency analysis and avoiding the related microstructure noise in option prices. Option prices are characterized by relatively large bid ask spreads and tick sizes (relative to their prices). Therefore, prices in option markets typically adjust more infrequently and only after sufficiently large changes in the underlying price or volatility. The volatility process is particularly noisy, as it is not traded directly and contains measurement error.

Delta and vega: A second important design choice relates to the calculation of trade deltas and vegas which are required for the order flow variables in Equations (2) and (3). We use the smile-consistent option pricing model of Heston (1993) to calculate deltas and vegas for each individual option trade. Our methodology for calculating these option sensitivities can be summarized as follows. First, we use the theoretical, functional relationship between the (observable) VIX index and the (unobservable) spot volatility to simultaneously calibrate the

¹⁰This filter drops less than 0.09% of the trades.

Heston model to intra-daily option data and the VIX index on the first day of our sample. Our approach avoids filtering techniques and high-dimensional optimization (Broadie, Chernov, and Johannes (2007), Christoffersen, Jacobs, and Mimouni (2010)) and is easy to extend to other pricing models or option markets. Second, we use the calibrated parameters from the first day of the sample and VIX index values on the next trading day to obtain the out-of-sample S&P 500 spot volatility. Third, equipped with estimates of the spot volatility and the structural parameters of the model, we then calculate deltas and vegas for all option trades on the second day using standard Fourier inversion techniques. The procedure is then repeated for all trading days in our sample. Our methodology allows for a conceptually easy construction of the trading exposure to the underlying S&P 500 (x^Δ) and the S&P 500 volatility (x^ν) using the definitions in Equations (2) and (3). We provide a more detailed description of our methodology in Appendix A.

Volatility process: We use the VIX index values to estimate the volatility process of the S&P 500 at a one minute frequency. We briefly describe our construction approach. We follow a standard procedure, and define the time- t VIX index with maturity τ by

$$VIX_t(\tau) = \left[\frac{2e^{r\tau}}{\tau} \int_{\mathbb{R}_0^+} \frac{O(t, k, t + \tau)}{k^2} dk \right]^{1/2} \quad (5)$$

where $O(t, K, T)$ denotes the time- t mid price of an out-of-the-money (OTM) option (with strike K and maturity T).¹¹ To approximate the integral in Equation (5), we first construct an option pricing function that is continuous in the strike K . We follow the literature and interpolate the implied volatilities of OTM options by a cubic polynomial and extrapolate the curve by fixing the implied volatilities of options beyond the traded strike range to the nearest available market implied volatility (for these procedures, see Broadie, Chernov, and Johannes (2007) and Carr and Wu (2009)). We then use a simple adaptive Gauss-Kronrod quadrature method to calculate the integral in Equation (5). We sample our option data at a one minute frequency to construct these data intra-daily. Appendix A describes the link between the VIX and spot volatility in more detail.

¹¹ Moneyness is defined relative to the forward price of the underlying.

4. Empirical Results

4.1 Summary Statistics

Individual option trades: Table 1 reports summary statistics of individual option trades. The full sample consists of more than 1.8 million trades. Column (1) shows that trades are generally large with an average size of \$93,300 which is about 20 times larger than a typical stock or ETF trade. Note that the dollar risk that changes hands in these trades is even larger, considering the embedded leverage options offer. The average SPX option trade is 51.4 contracts, where each contract consists of 100 options. Column (1) further shows that 41% of the option trades are calls (hence 59% are puts) and that 66.2% of the trades are buyer originated (therefore 33.8% are seller originated trades). The relatively strong buying demand in put options suggests that investors use options to hedge the downside risk in the S&P 500 index.

The absolute delta of put and call options is on average 0.248 with a standard deviation of 0.244, which suggests that there is considerable variation in the exposure option trades exhibit with respect to the underlying. Option trades also vary substantially in their volatility exposure with an average vega of 73.1 and a standard deviation of 50.2. This implies that an increase in (annualized) volatility from 15% to 16% leads to an increase in the price of a typical option by approximately $73.1 \times 0.01 = \$0.73$. Overall, these numbers indicate that investors actively trade options with a wide variety of characteristics, presumably to obtain varying exposures to the underlying and volatility. Column (1) also reveals that out-of-the-money options are traded relatively frequently, as the average moneyness (0.945) is less than one. Moneyness is defined as $F_{t,T}/K$ for puts and $K/F_{t,T}$ for calls where $F_{t,T}$ denotes the time- t forward price of the underlying. The average time to maturity is 0.118 years, or about 30 trading days.

The remaining columns in Table 1 report summary statistics for nine subsets of the data, double sorted by time to maturity (ttm) and moneyness. We create three buckets by ttm using the cutoff values of 30 and 90 days. We use cutoff values of 0.95 and 1.0 to group options into either deep out-of-the money, out-of-the-money or in-the money. While these cutoff values are arbitrary, our choice is driven to ensure that each subset includes a reasonable number of trades.

The same subsamples are used throughout the remainder of the paper. Each particular subset can be identified in the table by the corresponding average moneyness and ttm. As expected, the subgroups which contain near-the-money options yield the highest exposure to volatility (those in columns (3),(6) and (9)). Similarly, the subsets with highest moneyness, in columns (4), (7) and (10), yield the highest exposure to the underlying.¹²

Aggregated order flow variables: Table 2 provides summary statistics of all variables used in the estimation of Equation (4), including hourly order flow variables (in exposures of \$100 million),¹³ as well as the changes in the price of the underlying asset and its volatility. Our sample period is characterized by selling pressure in the SPY ETF, as the average of the order flow x^{spy} is -\$43 million per hour (with a standard deviation of \$432 million). The delta order flow constructed from option trades, measured in the same units, is positive with \$179 million per hour, and is twice as volatile with a standard deviation of \$864 million. This finding is in line with the summary statistics for individual trades, and confirms that investors in the SPX market trade in size.

The risk that changes hands through a one-standard deviation delta order flow trade is about the same as that of a vega order flow trade. To see this, note that the standard deviation of delta order flow is much larger than the standard deviation of vega order flow (\$3.92 versus \$0.16). However, as a pricing component, the volatility is a much riskier than the underlying.¹⁴ Together, the risk of holding a one-standard deviation delta flow trade for one hour is $3.92 \times 0.2 = 0.78\%$, while that of a vega order flow trade is $0.16 \times 5 = 0.8\%$.

¹²These subsets contain deep in the money calls (with a delta close to one) and deep in the money puts (with a delta just below zero). This explains why the average absolute delta of trades in these subsets ranges between 0.5 and 0.6.

¹³This unit is most convenient when estimating the structural VAR.

¹⁴The annualized volatility level is on average 11.5% with an hourly standard deviation of 0.58 percentage points, which gives an hourly standard deviation of volatility *return* of $5.0\% = 0.58/11.5 \times 100$. The underlying has a spot price of \$1,930 on average and its difference d^u has a standard deviation of \$3.92 per hour, giving a standard deviation of the underlying *return* of $0.2\% = 3.92/1,930 \times 100$.

4.2 Price Impact Model

We estimate the VAR model in Equation (4) with two lags as suggested by the Akaike Information Criterion. Following Hasbrouck (1991), we assume the trading process restarts at the beginning of each trading day, and therefore we set all initial lagged values to zero.

Table 3 provides the long run impulse response functions (IRFs) of the model. The table contains the five-by-five matrix of IRFs, which shows the impact of an impulse in one variable (in columns) on the cumulative impact on all other variables (in rows). We allow each shock to iterate through the system for four periods. The table also shows the size of each shock in the first column. Table 3 offers two main results. First, column (6) shows that a one standard deviation shock in x^v increases d^v by 0.07. This translates into a volatility increase from the sample average of 11.5% to 11.57%, and corresponds to 0.14 standard deviations of d^v . This magnitude is economically sizeable because it is caused by vega order flow. The remaining variation in volatility is driven by public information arrival and the other order flows. The second main result of Table 3 is shown in column (5), where the long-run impact of a shock in x^Δ on d^u is 98.13 cents. This can be interpreted as a change in the average value of the underlying from \$1,930 to \$1,930.98 which corresponds to an increase of 5.1 basis points. The effect is economically large as it corresponds to 50% of the standard deviation of d^u at the hourly frequency (\$1.90). This finding for the S&P 500 is consistent with the results for equity options in Muravyev (2016). Interestingly, the impact of an order flow shock in the ETF (x^{spy}) is larger with \$1.71 per standard deviation (see column (4)). Also note that the size of ETF shocks are much smaller than the size of delta flow shocks: \$420 million vs \$823 million. Accordingly, the *per-dollar* price impact of an ETF shock is 3.4 times larger than when the same exposure is obtained through options.¹⁵

Table 3 provides several additional findings. The impact of x^{spy} on d^v is negative and large, which is driven by the leverage effect (i.e., the negative correlation between d^v and d^u). The mechanism is that x^{spy} increases d^u , which simultaneously decreases d^v . This correlation occurs contemporaneously and cannot be disentangled by the VAR. The same argument explains why x^Δ

¹⁵The dollar impact of an ETF order flow shock is \$1.71/4.20, which is 3.4 times larger than that for options: \$0.98/8.23.

negatively affects d^v , and x^ν negatively affects d^u .¹⁶ Further, the coefficients on the diagonal of Table 3 reveal the long-term impact of an impulse on the same variable. All variables are positively autocorrelated, because the long-term impacts are greater than the size of the structural shocks (shown in the first column). In addition, x^Δ and x^ν are uncorrelated in the long-run as a shock in one does not affect the other. Indeed, x^ν and x^Δ mechanically have a positive correlation for call option trades and a negative correlation for put option trades — on average, the two opposing effects seem to cancel out. Lastly, ETF flow x^{spy} significantly causes delta flow in the long-run, but not the other way around. This suggests that x^{spy} is leading and delta flow following.

The plots in Figure 1 reveal the cumulative impacts over time. For brevity, we only show the shocks of the order flow variables on the changes in the price and volatility. The figure reveals that the majority of the effects occurs in the first hour.

Table 4 contains the regression coefficients of the VAR model estimated by OLS. The omitted coefficients in the columns reveal the ordering of the equations in the structural model (which in turn identifies the structural residuals). The results of the IRFs are clearly visible in the coefficients as well. Furthermore, the R-Squared of d^v is 0.25, meaning a quarter of its variation can be explained by the three contemporaneous order flows and all the lagged variables. The R-Squared of d^u in column (5) is very high with 0.766, but this is driven by d^v which appears contemporaneously in the regression and picks up the leverage effect (with a t -stat of 32.2).¹⁷ Furthermore, the positive contemporaneous coefficient of x^Δ on x^{spy} (0.086) suggests that traders obtain exposure to the underlying by trading simultaneously options and the ETF. The coefficient would be negative if traders were to hedge positions in options by trading the ETF. The table also confirms the aforementioned positive autocorrelations in the order flow variables.

These results have several important implications. First, our findings suggest that traders who are privately informed about the “fundamental” volatility level trade options to exploit this information. Option market makers in turn, aware of these informed traders, observe the order flows and update the prices of *all* options to reflect this volatility information. It is important

¹⁶The intuition is that x^Δ positively predicts d^u , and therefore negatively d^v because of the negative correlation between d^u and d^v . Similarly, x^ν positively predicts d^v , and therefore negatively d^u .

¹⁷Indeed, the underlying and volatility process are simultaneously determined and have a strong negative correlation of -0.858 at the hourly frequency.

to emphasize that the volatility process is extracted from all options in the cross section of strike prices. This suggests that we indeed measure fundamental volatility information, and not a transitory impact that could be caused by a temporary trading demand in a particular option. That is, it seems unlikely that a transitory demand pressure in a single option would affect prices of all options and therefore the volatility process.

A second implication is that we confirm that delta order flow exposure indeed affects the price of the S&P 500. While this may seem obvious at first, the empirical literature provides many contradicting findings. Several papers find no meaningful impact of option order flow on the underlying (e.g., [Stephan and Whaley \(1990\)](#), [Chan, Chung, and Fong \(2002\)](#), [Chordia, Kurov, Muravyev, and Subrahmanyam \(2019\)](#)), whereas others do (e.g., [Pan and Poteshman \(2006\)](#), [Muravyev \(2016\)](#)).

A third implication of our findings is that cross-option price impacts can be very significant to large investors who wish to trade multiple options. Indeed, a trade in one option affects the prices of others through its impact on the underlying and volatility. Accordingly, we should not analyze the liquidity in the order books of individual options, but rather estimate it through a joint system. This implication is consistent with the literature on market fragmentation, where the same asset is traded on multiple exchanges. In our setting, highly correlated options (rather than the same asset) are traded in different limit order books.

And fourth, our finding that ETF order flow has a 3.4 times higher per dollar price impact than delta flow suggests a degree of market segmentation between SPX option markets and the SPY ETF. Option trades are much larger in terms of dollar volume and risk, which suggests that investors who prefer to trade in size use options. In contrast, the ETF market attracts relatively small and informed trades. The low price impact of delta flow is consistent with [Muravyev and Pearson \(2019\)](#), who find that effective option bid-ask spreads that take into account trade execution timing are three times smaller than conventionally measured bid-ask spreads. That is, option market exhibit a higher liquidity compared to what the conventional literature reports, which is consistent with our finding that option markets have a low price impact.

4.3 Testing the Imposed Structure

In this section we evaluate two crucial assumptions our structural model imposes on the data. First, we test whether the price impacts of delta and vega flow are the same when the order flow variables are constructed from trades in different option subsets. Second, we investigate whether summarizing the cross section of option price changes with a two-dimensional process leads to any substantial loss in information.

Price impacts for different types of options: We construct the delta and vega order flows separately for trades in the nine subsets of options introduced in Section 4.1, which are double sorted by time to maturity and moneyness. We estimate the equations in the VAR by OLS using the original delta and vega flows, and add the flows of the individual subgroups. For each regression we perform a single Wald test, testing whether the cumulative effect (the contemporaneous term and the two lags) of the nine subgroups are jointly equal to zero.¹⁸

The results of the Wald tests are as follows. For the d^v equation, the p -value of the joint test for the vega flows is 0.262, so we fail to reject equality of the vega flows across subgroups. Similarly, the joint test of vega flow in the d^u equation is 0.694. Turning to the joint test for delta flows, we do reject equality in both the d^v and d^u regression. Further inspection reveals that the rejection is caused by the coefficient of the subgroup of options with low moneyness and a short time to maturity: the delta flow from this group has a significantly higher impact on d^u and a more negative impact on d^v . The trades in this group have a very low delta (because they are out of the money), and therefore the delta flow takes small values. The market seems to respond stronger to these trades.

Information loss using a two-dimensional process: The five equation VAR model is designed to summarize the information captured in the large state space of options prices and order flows across strike prices, expiration dates and puts and calls. To evaluate the performance of this data reduction technique we follow Bakshi, Cao, and Chen (2000), and test the model

¹⁸We have nine subgroups, but by adding the aggregate delta and vega flows we lose one subgroup which becomes the base case. The test is not affected by which group becomes the base case, because we test whether the cumulative coefficients of all groups are identical. With two lags in the system, we jointly test whether the 8 sums (of three coefficients) of the delta flows (or vega flows) are equal to zero. We perform the test for the d^v and d^u equations in the VAR.

in Equation (1) by regressing the hourly price change of individual options, with characteristics $n=\{Strike, Expiration\ date, Put-call\ identifier\}$, on the price and volatility changes according to the following specification:

$$d_{n,t}^o = \beta_0 + \sum_{l=0}^2 \beta_{l,1} \Delta_{n,t-l-1} d_{t-l}^s + \sum_{l=0}^2 \beta_{l,2} \nu_{n,t-l-1} d_{t-l}^v + \varepsilon_{n,t}, \quad (6)$$

where $d_{n,t}^o$ is the change in the mid quote of option n , and the terms $\Delta_{n,t} d_t^s$ and $\nu_{n,t} d_t^v$ represent the options exposure (delta or vega) multiplied by the factor (change in underlying price or change in volatility). This analysis uses data of all individual options (one observation per option-day-hour), whereas the VAR used one observation per day-hour. For consistency with the VAR model we include two lags in the regression. If the structure imposed by a two-factor option pricing model is correct, the regression should yield a) an R-Squared of one, and b) coefficients $\beta_0 = 0$, $\sum_{l=0}^2 \beta_{l,1} = 1$, and $\sum_{l=0}^2 \beta_{l,2} = 1$. Indeed, in this theoretical case the empirical specification would perfectly explain the actual changes in option prices.

Table 5 provides our estimation results. Column (1) shows the full sample regression. We obtain an R-Squared of 99.2%, which suggests that our model does an excellent job at summarizing the information content in option price changes. We also find that $\sum_{l=0}^2 \beta_{l,1} = 0.997$ and $\sum_{l=0}^2 \beta_{l,2} = 0.899$, which both are close to one from an economical point of view. This also confirms that option returns are strongly affected by changes in volatility, and that options are useful tools to speculate on changes in volatility. The t -statistic on coefficient $\Delta \times d^s$ is extremely large (5,551) due to the high R-Squared and the sheer size of the data (851,051 observations). For this reason, we do reject equality to one for both $\sum_{l=0}^2 \beta_{l,1}$ and $\sum_{l=0}^2 \beta_{l,2}$ (the t -statistics are 14.6 and 37.3, respectively). Column (1) further shows that the lagged coefficients are smaller, but still statistically significant. This motivates the use of the lags in the system. Compared to Bakshi, Cao, and Chen (2000), the two factor model works much better with more recent data because markets have become more efficient. Using SPX option data from 1994, they find coefficients of $\beta_1 = 0.80$, $\beta_2 = 0.41$, and an R-Squared of 59%.

We repeat the exercise for the nine subsets of options sorted by time to maturity and money-

ness. In general, the model works very well. We see it performs slightly weaker for options with a low moneyness and short time to maturity (column 2). In this case, the $\sum_{l=0}^2 \beta_{l,1} = 0.612$ and $\sum_{l=0}^2 \beta_{l,2} = 1.35$. These coefficients are likely affected by the model misspecification of the Heston model, which, e.g., does not model any discontinuous jump moves in the underlying price equation. In particular, it has been shown that for pricing short time to maturity options modeling a jump component yields better pricing performance compared to a pure stochastic volatility model (see Eraker (2004)). Further, this subset contains options with very low prices, and we know that the microstructure noise is more severe here as tick sizes are relatively larger. The model works better for the remaining columns, which all have an R-Squared exceeding 92%.

From an economical standpoint the imposed structure fits the data well. However, the rejection of equality to one for $\sum_{l=0}^2 \beta_{l,1}$ and $\sum_{l=0}^2 \beta_{l,2}$ means that actual option price changes differ from what our model predicts. This implies that either the deltas and vegas contain errors; or options price changes have transitory components; or that the true data generating process for option prices contains additional factors (see e.g., Christoffersen, Heston, and Jacobs (2009) or Bardgett, Gouriéroux, and Leippold (2019)). As a consequence, there is some bias in the delta and vega order flows used in the VAR model. This issue could be tackled by using a more advanced option pricing model, but this is beyond the scope of this paper. However, we see no economic channel how any misspecification of the two-factor structure would alter the findings in the VAR. In fact, any misspecification would most likely bias the results against finding evidence of informed trading.

5. Higher Frequencies

We have shown that at the one-hour frequency delta and vega order flows predict changes in the underlying and volatility. Is it possible to extend these findings to higher frequencies? To address this question, we sample our data at the half-hour and one-minute level and repeat the analyses.

Estimating the VAR model for higher frequencies such as one-minute intervals poses a non-trivial challenge since a much larger number of lags needs to be added to the VAR equations to

cover a comparable time horizon as for the one-hour frequency. A large number of lags implies a large number of parameters, which in turn makes estimation procedures unstable. To deal with this issue we follow the procedure proposed in [Hasbrouck \(2018\)](#) which uses polynomial distributed lag (PDL) functions to decrease the number of model parameters. In this approach, the time horizon covered by the lags is divided into subintervals, and for each interval a PDL function is fitted. This procedure reduces the number of parameters significantly by imposing a structure on the lagged coefficients. After estimating the parameters of the PDL function the parameters for the individual lags and impulse response functions can be calculated.

Our main finding is that the VAR analysis provides weaker results at the half-hour frequency, and insignificant results at the one-minute frequency. Specifically, the coefficients and IRFs of delta flow and vega flow on changes in the underlying and volatility shrink, and they turn insignificant at the one-minute frequency (see Figures 2 and 3, respectively). After careful investigation we conclude that at higher frequencies microstructure noise becomes too severe, weakening all effects of interest. Options have large tick sizes and bid-ask spreads, which prevents prices to adjust smoothly to small changes in the underlying asset or the volatility. Only at sufficiently low frequencies the changes in the two factors are large enough (relative to bid-ask spreads) to induce option price changes.

We provide two analyses to reveal the effect of microstructure noise on the estimates. First, Table 6 shows that the bid-ask spreads and tick sizes of options are large compared to the standard deviation of option price changes at high frequency. The tick size ranges from \$0.05 to \$0.10 (depending on the option price level).¹⁹ The bid-ask spreads however, calculated as the median across option categorized into eight subsets by option price level, range from \$0.35 for options priced under a dollar to \$1.56 for options priced between \$20 and \$40. This means the spreads are typically between 7 to 15 ticks. The spread values are similar to the standard deviation of price changes at the hourly level, which range from \$0.14 to \$1.45 for the same subsets of options. At the one-minute frequency, however, the standard deviations of price changes range between \$0.05 and \$0.17, which is about 7 to 10 times smaller than the bid-ask spread. This implies that

¹⁹Currently, SPX options have a tick size of \$0.05 when the price is below \$3.00, and \$0.10 otherwise. Source: <http://www.cboe.com>

at the one-minute level, the changes in the two price factors are small and thus the implied option price changes are small. Such changes often fall within the bid and ask quotes, which therefore are unlikely to update frequently.

The direct consequence of the microstructure noise is that the two-factor structure we impose has weak explanatory power at the highest frequencies. In Table 7 we repeat the analysis of Section 4.3 at the one-minute frequency, and find that the two factors explain no more than 47.9% of the variation in the full sample. Moreover, the coefficients on the terms Δd^s and νd^v (summed over the lags) lie much further away from 1. These issues are worse for the subsets of the options with low moneyness (columns 2, 5 and 8), where the R-Squared is only 42.1%, 28.8% and 19.1%, respectively. The two-factor structure is in essence a data reduction technique that summarizes the information in many option prices and order flows by a handful of components. This structure appears to break down at the highest frequencies.

The poor performance of the two-factor model at the high frequency also implies weaker results of the VAR model. Quoted option prices do not update frequently and thus will not reflect order flow information about the underlying or volatility. In addition, the volatility process is extracted from option prices, which will also update more slowly. A second issue is that the leverage effect seems to become stronger at higher frequencies. This makes it more complicated for the VAR model to distinguish between the impact of flows on the underlying and the volatility component. Given that vega flow is ordered last in the VAR, any contemporaneous correlation between vega flow and SPY flow (or vega flow and delta flow) will be attributed to the latter, leaving less explanatory power for vega flow. This effect can be seen by comparing the minute-frequency analysis in Figure 3, where, compared to the hourly analysis in Figure 1, the x^{spy} gets a stronger predictive power at the expense of x^ν and x^Δ . Overall, we find that the hourly frequency used throughout the paper is the best compromise between a high frequency analysis and avoiding the adverse effects of market microstructure noise in option data.

6. Robustness Analyses

We have performed a number of robustness analyses which we summarize in this section. Detailed results can be found in the Web Appendix.

Ordering of the equations. An important but somewhat arbitrary choice is the ordering of the equations in the structural VAR. The ordering determines the direction of the contemporaneous correlation between two variables: only one is allowed to affect the other, but not the reverse. Without this structure the system is not identified. The ordering in turn determines the structural innovations, the regression coefficients, and the impulse response functions. While five equations allow for 120 possible combinations, there are two alternative orderings that deserve further attention. First, we estimate a version of our model where we reorder d^u and d^v which switches the direction of the leverage effect. In this case, d^v strongly causes d^u , and the same holds for contemporaneous order flows. The long-run impulse response functions, however, are nearly identical to our original estimates (see Figure 1 in the Web Appendix). And second, we estimate a version of our model where we put x^{spy} before x^Δ and x^ν (see Figure 2 in the Web Appendix). In this case, the information contemporaneously captured in x^{spy} and x^Δ is attributed to the latter. Accordingly, the coefficient of x^Δ on d^u is estimated more precisely (a higher t -stat). The long-run effects however are very similar to the main specification, both in terms of economic magnitudes and significance levels.

The forward price. In the main specification we use the ETF SPY price to proxy for the value of the underlying asset. An alternative is to extract the forward price from the option data, which is noisier but not confounded by dividend yields paid before option expiration.²⁰ In a robustness check, we estimate the VAR using the forward price as the underlying instead. These results are virtually identical which is reassuring. Our preferred specification however uses the ETF price which typically leads in price discovery (Hasbrouck, 2003). At the one-hour frequency,

²⁰We back out the underlying price process from option prices as follows. For each t on a one-minute grid, we collect every put/call pair with identical strike price and identical maturity date and use market mid quotes to solve put-call parity for the unobservable futures prices. At a given time t and for a fixed option maturity T , we average the implied futures prices over all available strikes for which a put and a call price is available to obtain the option implied futures price $F_{t,T}$. We use linearly-interpolated rates from the OptionMetrics zerocurve file as a proxy for the risk-free rate of interest. We then average the changes over all short-term futures in our sample to calculate d_t^u .

however, this channel is of no concern.

Price changes versus returns. We estimate the VAR with the return on the underlying and volatility, instead of price differences. After appropriately scaling the coefficients, the results are virtually identical to those in the main specification. We chose to report the version with price differences, as it corresponds more naturally to the analysis in Section 4.3.

The role of the SPY ETF order flow. We estimate a four equation version of the VAR model, where we drop the SPY ETF order flow equation from the model. In this case, results are generally stronger in terms of statistical significance (although the point estimates of the coefficients are similar). Not only do we estimate fewer parameters, also note that the ETF order flow is correlated with option order flows. When adding ETF flow in the main specification of the model, it absorbs part of the predictive power.

Limitations: The analysis is based on data of SPX options and the SPY ETF. In reality, investors can trade other vehicles to obtain exposure to the S&P 500 (through the S&P 500 future for example) or its volatility (through options on the VIX or options on the SPY for example). We do not have these data and thus are unable to investigate to which extent investors use such sources to trade on their information. Nonetheless, the main contribution of the paper is the model, which is flexible and allows for easy integration of other order flow sources. One approach is to calculate the delta and vega order flows in these additional assets and sum them with the current variables. An alternative is to incorporate separate equations for these sources, which allows for tests of differences of coefficients to determine which order flows have higher price impacts.

A conceptual limitation of the model is caused by the leverage effect, i.e., the negative correlation between changes in the underlying and the volatility. This correlation occurs simultaneously between the variables, and the VAR model cannot identify the causality. This issue complicates interpretation of some coefficients, because for example the negative effect of x^{spy} on d^v is likely caused by the positive impact of x^{spy} on d^u .

7. Conclusion

We offer a novel framework to estimate the liquidity of the aggregate options market. To our knowledge, we are the first to propose a single model that captures trading in potentially hundreds of options, with differing strike prices, expiration dates and types (put or call). We impose economic structure on the data using theoretical option pricing models, which takes into account the high cross-option correlations in returns, order flows, and liquidity.

The empirical results using S&P 500 (SPX) options are consistent with classical models of informed trading. We find that order flow exposures to the underlying (or volatility) positively predict changes in the price (and volatility), consistent with strategic traders using options to speculate on changes in either component. As a result, order flow in one option affects the underlying and volatility, and therefore indirectly the prices of all other options. This result is key for large traders concerned about transaction costs when trading multiple options.

We believe the results are driven by asymmetric information (informed trading), and not by transitory demand pressures (inventory concerns). We use the price of the underlying asset and extract the volatility from the cross section of options. It is unlikely that a transitory demand pressure in one option moves all option prices simultaneously. Further, our VAR system contains several lags, such that the price changes are permanent up to this lag number.

We see an important analogy with the literature on equity market fragmentation, where the same asset is traded on multiple trading venues. In the current setting, highly correlated options are traded in multiple limit order books. The economic forces of informed trading strategies across markets (or options) are similar, including the resulting correlations in order flows, returns, and liquidity.

Conceptually, our approach can easily be extended to options on individual stocks, or to other markets where groups of highly correlated assets are traded. For example, for the corporate bond market a term structure model could be used to estimate the bond pricing factors. Then, each bond trade gives exposure to these factors, which could be used in a similar structural VAR model with factor returns and order flow exposures.

A. Appendix: Heston Model Details

Model definition. The [Heston \(1993\)](#) model has become the most important benchmark in the option pricing literature. Its main theoretical advantage stems from the mathematical tractability of its characteristic function (see [Duffie, Pan, and Singleton \(2000\)](#)). Under the model assumptions, the S&P 500 index S and its volatility v are described by the following stochastic differential equations under the risk-neutral measure $\mathbb{Q}$:

$$dS_t = (r - q)S_t dt + S_t v_t dW_t^1 \quad (\text{A.7})$$

$$dv_t^2 = \kappa^{\mathbb{Q}} (\theta^{\mathbb{Q}} - v_t^2) dt + \sigma v_t (\rho dW_t^1 + \sqrt{1 - \rho^2} dW_t^2) \quad (\text{A.8})$$

where W^1 and W^2 are standard Brownian motions under $\mathbb{Q}$. Structural parameters are given by $\kappa^{\mathbb{Q}} > 0$, $\theta^{\mathbb{Q}} > 0$, $\sigma > 0$ and $\rho \in [-1, 1]$. The risk-free rate is denoted r and q is the continuous dividend yield.

Due to the affine model structure, European call and put prices can be calculated by standard Fourier inversion methods (see below) and we denote $P(t, S_t, v_t, K, T, \omega)$ the time- t price of a put or call option ($\omega = 1$ for a call, $\omega = -1$ for a put) with strike K and maturity $T > t$. We collect all structural parameters and the risk-free rate in the vector Θ but drop the dependence of option prices on the parameter vector for notational simplicity. It follows from Ito's Lemma that

$$dP(t, S_t, v_t, K, T, \omega) = \frac{\partial P(t, S_t, v_t, K, T, \omega)}{\partial S_t} dS_t + \frac{\partial P(t, S_t, v_t, K, T, \omega)}{\partial v_t} dv_t + dt\text{-terms} \quad (\text{A.9})$$

Due to the analytical tractability of the option pricing function P , the first partial derivatives with respect to S and v can also be calculated analytically and we denote these as:

$$\Delta(t, S_t, v_t, K, T, \omega) \equiv \frac{\partial P(t, S_t, v_t, K, T, \omega)}{\partial S_t} \quad (\text{A.10})$$

$$\nu(t, S_t, v_t, K, T, \omega) \equiv \frac{\partial P(t, S_t, v_t, K, T, \omega)}{\partial v_t}. \quad (\text{A.11})$$

To relate changes in option prices to observable variables, we exploit the theoretical relation

between the spot volatility and the VIX index. It is straightforward to show that under our model assumptions, the squared VIX index is a linear function of spot variance, i.e. $VIX_t^2(\tau) = a(\Theta, \tau) + b(\Theta, \tau)v_t^2$ where a and b are functions of the structural model parameters Θ and the maturity of the options used to construct the VIX index (denoted τ , i.e. 30 days for the standard index published by CBOE). Using these theoretical results, given a parameter set Θ , option delta and vega can also be calculated by Fourier inversion.

Option pricing formulae. Duffie, Pan, and Singleton (2000) show that the generalized characteristic function for affine jump-diffusion processes is exponentially affine in the state variables $\log S_t$ and v_t^2 . For the Heston model, the logarithm of the characteristic function under the risk neutral measure $\mathbb{Q}$ is given by

$$\log \Psi^{\mathbb{Q}}(u, \tau, S_t, v_t) \equiv \log E^{\mathbb{Q}} \left[e^{iu \log S_{t+\tau}} \middle| \mathcal{F}_t \right] = \mathcal{A}(u, \tau) + \mathcal{B}(u, \tau)v_t^2 + u \log S_t,$$

where $\mathcal{A}$ and $\mathcal{B}$ are complex-valued functions, $i = \sqrt{-1}$ and $E^{\mathbb{Q}}[\cdot | \mathcal{F}_t]$ denotes the risk-neutral $\mathcal{F}_t$ -conditional expectation. For expositional clarity the dependence of all functions on the parameter set of the model Θ is suppressed. Following Bates (2006), the price of a European call option is given by

$$P(t, S_t, v_t, K, T, \omega = 1) = e^{-r\tau} F_{t,T} - e^{-r(T-t)} K \left(\frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \Re \left[\frac{e^{-iu \log K} \Psi^{\mathbb{Q}}(u, T-t, S_t, v_t)}{iu(1-iu)} \right] du \right).$$

where $F_{t,T} = e^{(r-q)(T-t)} S_t$ and $\Re$ denotes the real part of a complex number. Partial derivatives of the call price are given by

$$\frac{\partial P(t, S_t, v_t, K, T, \omega = 1)}{\partial S_t} = e^{-q(T-t)} - \frac{e^{-r(T-t)} K}{\pi} \int_0^{\infty} \Re \left[\frac{iu}{S_t} \frac{e^{-iu \log K} \Psi^{\mathbb{Q}}(u, T-t, S_t, v_t)}{iu(1-iu)} \right] du$$

and

$$\frac{\partial P(t, S_t, v_t, K, T, \omega = 1)}{\partial v_t} = -\frac{2v_t \times e^{-r(T-t)} K}{\pi} \int_0^{\infty} \Re \left[\frac{\mathcal{B}(u, T-t) e^{-iu \log K} \Psi^{\mathbb{Q}}(u, T-t, S_t, v_t)}{iu(1-iu)} \right] du.$$

The theoretical value of the VIX index with maturity $\tau_v = 30/365$ can be recovered from the characteristic function. One can show that for the Heston model

$$VIX_t(\tau) = \sqrt{\theta^{\mathbb{Q}} \left(1 - \frac{1 - e^{\kappa^{\mathbb{Q}}\tau}}{\tau \kappa^{\mathbb{Q}}} \right) + \frac{1 - e^{\kappa^{\mathbb{Q}}\tau}}{\tau \kappa^{\mathbb{Q}}} v_t^2}. \quad (\text{A.12})$$

Empirical estimation. In order to compute the delta and vega for all option transactions in our dataset, we need to calibrate the Heston model to market data. There is no standard methodology in the literature regarding the estimation of model parameters of option pricing models. We adopt a simple calibration procedure and estimate model parameters for every trading day in our sample using intradaily option quotes. To this end, we minimize the mean-root-squared-error of all available OTM options with maturities from 7 to 180 days as follows:

$$RMSE_t = \arg \min_{\Theta} \sqrt{\frac{1}{N_t} \sum_{i=1, \dots, N_t} (P^{ma}(t_i, K_i, T_i, \omega_i) - P(t_i, S_{t_i}, v_{t_i}, K_i, T_i, \omega_i, \Theta))^2}, \quad (\text{A.13})$$

where N_t is the number of available option quotes on day t , t_i are set to minutely intervals from 9:45am to 15:45pm (on day t) and $P^{ma}(t_i, K_i, T_i, \omega_i)$ denotes the market mid-quote of option i . We use the shortest option maturity to invert the theoretical VIX formula above to obtain v_t . To limit the number of option contracts in our calibration, we restrict the calibration to the most liquid contracts, OTM contracts and contracts with short to medium maturity. For the calculation of deltas and vegas we use calibrated parameters from the previous trading day.

Our calibration procedure has two distinct features. First, our methodology circumvents the problem of filtering the latent variance process, as we calibrate the model to both option prices and the VIX index simultaneously. Alternative approaches such as those in Broadie, Chernov, and Johannes (2007) or Christoffersen, Jacobs, and Mimouni (2010) rely on filtering techniques or the calibration of v_t which leads to additional complexity in the calibration algorithm. And second, our recalibration allows us to be robust to changing market dynamics. While the Heston model may be rejected because of the imposed structure (see for instance Broadie, Chernov, and Johannes (2007) or Christoffersen, Jacobs, and Mimouni (2010)), our recalibration ensures that empirical results are not particularly sensitive to possible model misspecification. To calculate

deltas and vegas for option transactions, we use parameters estimated using option data on the previous day, so our procedure is out-of-sample. While parameters in the calibration may change over time, our assumption is that the Heston model provides a reasonable way of separating price from volatility risk using model calibrations from the previous trading day. Following the empirical set-up in Bakshi, Cao, and Chen (2000), we provide strong empirical evidence in Table 5 that this approach covers more than 98% of the variation in option prices in our sample. Hence, Equation (A.9) in combination with our Heston model implementation provides a highly accurate description of option market prices, and offers the main benefit of a reduction of variables to only price and volatility risk.

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Table 1 Summary statistics SPX option trades on S&P500.

The table shows the mean and standard deviation (in brackets) of option trade variables. The first column reports results for the full sample, and the remaining columns those for nine subsets of the data double sorted by moneyness and time to maturity. Each subsets can be identified by the rows showing the average Moneyness and Time to Maturity. Most variables are self explanatory. The vegas are calculated with respect to the underlying volatility. The trade direction equals 1 if the trade is originated with a market buy order, and -1 if it is a market sell order. The call indicator equals 1 if the trade is in a call option, and zero if it is in a put option.

	Full Sample	Subsets								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Volume (\$1,000)	93.3 (343)	6.73 (40)	37.7 (150)	137 (464)	28.8 (149)	133 (352)	231 (585)	98.6 (320)	306 (583)	389 (919)
Volume (#contracts)	51.41 (124.9)	50.56 (134.5)	44.16 (115.1)	41.37 (104.1)	52.30 (139.3)	56.59 (125.7)	47.49 (99.02)	75.58 (159.7)	70.39 (122.8)	50.46 (97.71)
Price (\$)	18.02 (38.05)	1.085 (1.358)	7.268 (8.996)	39.29 (68.71)	4.197 (4.751)	22.12 (16.84)	54.26 (71.71)	9.139 (10.06)	43.57 (15.76)	89.03 (114.3)
Option abs(Δ)	0.248 (0.244)	0.0135 (0.0188)	0.217 (0.170)	0.652 (0.181)	0.0424 (0.0414)	0.322 (0.177)	0.572 (0.168)	0.0682 (0.0727)	0.406 (0.152)	0.565 (0.177)
Option ν	73.08 (50.21)	11.63 (15.47)	75.62 (39.58)	103.8 (33.95)	32.87 (28.23)	108.9 (37.53)	127.0 (31.97)	35.14 (29.35)	97.50 (23.47)	97.04 (28.91)
Trade is buy (1,0)	0.662 (0.473)	0.587 (0.492)	0.604 (0.489)	0.635 (0.481)	0.608 (0.488)	0.640 (0.480)	0.678 (0.467)	0.624 (0.484)	0.633 (0.482)	0.620 (0.485)
Call indicator (1,0)	0.411 (0.492)	0.0928 (0.290)	0.528 (0.499)	0.579 (0.494)	0.194 (0.396)	0.559 (0.496)	0.521 (0.500)	0.222 (0.416)	0.531 (0.499)	0.493 (0.500)
Moneyness	0.945 (0.0967)	0.878 (0.0698)	0.979 (0.0139)	1.017 (0.106)	0.860 (0.0973)	0.980 (0.0164)	1.015 (0.109)	0.823 (0.114)	0.982 (0.0164)	1.031 (0.188)
Time to Maturity (years)	0.118 (0.0853)	0.0500 (0.0201)	0.0460 (0.0196)	0.0482 (0.0194)	0.142 (0.0427)	0.139 (0.0433)	0.151 (0.0460)	0.330 (0.0697)	0.302 (0.0626)	0.293 (0.0586)
Number of trades	1,804,397	213,636	434,864	109,737	333,048	387,633	165,202	74,815	53,324	32,138

Table 2 Summary statistics on changes in price and volatility of the S&P500 and their respective order flows.

We calculate statistics for the sample of SPX options and the SPY ETF for 249 trading days in 2014. Order flow is defined as buyer originated minus seller originated trading volume and expressed in dollars. The first row shows statistics on the net order flow in the SPY ETF (in \$100s of millions). For the next two rows, we decompose each SPX option trade into exposure to the underlying S&P500 (x^Δ), based on the options delta, and exposure to the S&P500 volatility (x^ν), based on the options vega. The order flow exposures are then aggregated over all trades across options with different strike prices, expiration dates, and puts and calls. We further report statistics on the differences in the SPY ETF price (d^u) and the S&P500 volatility process (d^v) extracted from option prices. The volatility and the option deltas and vegas are estimated with a Heston stochastic volatility model using the cross section of option prices.

	Mean	StDev	p10	p25	p50	p75	p90	N
x^{spy} (\$100 million)	-0.43	4.32	-5.24	-2.75	-0.26	1.86	4.26	1,494
x^Δ (\$100 million)	1.79	8.64	-7.84	-2.65	1.51	6.07	11.57	1,494
x^ν (\$100 million)	0.11	0.16	-0.02	0.02	0.07	0.16	0.28	1,494
d^u (cents)	5.86	392.91	-450.00	-189.99	14.99	220.00	439.99	1,494
d^v (percentage points)	-0.01	0.58	-0.59	-0.26	-0.03	0.22	0.59	1,494

Table 3 Long-run impulse response functions VAR model.

This table shows the long-run effects of the impulse response functions (IRFs) of the VAR model estimated by Equation (4). Long-run is defined as four periods (trading hours). The first column shows the standard deviation of the structural residual of each equation, which represents the size of the shocks in the IRF. The next columns show the long-run impact of a one-standard deviation impulse to each of the variables. The superscripts ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Response:	$\sigma(\epsilon)$ (1)	Impulse:				
		d^u (2)	d^v (3)	x^{spy} (4)	x^Δ (5)	x^ν (6)
d^u (cents)	189.93	215.70*** (11.28)	-277.10*** (14.14)	171.00*** (8.06)	98.13*** (4.41)	-55.64*** (2.59)
d^v (percentage points)	0.50	0.00 (0.09)	0.52*** (18.75)	-0.23*** (7.45)	-0.11*** (3.31)	0.07** (2.36)
x^{spy} (\$100 million)	4.20	-0.38* (1.67)	0.62*** (2.86)	4.91*** (19.99)	0.44 (1.62)	-0.28 (1.10)
x^Δ (\$100 million)	8.23	2.46*** (5.17)	-1.28*** (2.75)	1.03** (2.11)	9.87*** (18.21)	-0.16 (0.29)
x^ν (\$100 million)	0.16	0.01 (0.62)	0.02** (1.99)	-0.02* (1.73)	-0.01 (0.51)	0.21*** (20.60)

Table 4 Regression coefficients VAR model.

This table shows the coefficients of the VAR system estimated from Equation (4). The system considers five endogenous variables reported in columns (1) to (5) respectively: the net dollar option order flow exposure to the volatility component (x^ν), to the underlying component (x^Δ), and net SPY ETF order flow (x^{spy}); and the difference of the volatility process VIX (d^v) and the price of the underlying measured by the SPY ETF (d^u). The ordering of the columns corresponds with the ordering of the equations in the VAR model, and the coefficients set to missing identify the structural shocks. The letters L in the independent variable names represent the lag-operator. Inference is based on Newey-West standard errors with two lags. The superscripts ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1) x^ν	(2) x^Δ	(3) x^{spy}	(4) d^v	(5) d^u
x^ν	.	0.039 (0.0)	-2.137** (-2.1)	0.457*** (3.9)	37.547 (1.0)
L1 x^ν	0.205*** (3.1)	0.737 (0.4)	0.410 (0.5)	-0.120 (-0.9)	-57.625 (-1.5)
L2 x^ν	0.026 (0.7)	0.035 (0.0)	0.131 (0.2)	-0.072 (-0.8)	-41.673 (-1.1)
x^Δ	.	.	0.086*** (6.0)	-0.012*** (-6.8)	2.735*** (3.8)
L1 x^Δ	0.001 (0.5)	0.178*** (4.3)	-0.031** (-2.0)	0.002 (0.7)	0.446 (0.6)
L2 x^Δ	-0.001 (-0.8)	-0.076** (-2.3)	0.001 (0.0)	0.003 (1.5)	0.263 (0.4)
x^{spy}	.	.	.	-0.056*** (-10.6)	14.799*** (8.5)
L1 x^{spy}	-0.002 (-1.3)	-0.002 (-0.0)	0.165*** (4.0)	0.011*** (2.7)	-1.441 (-1.0)
L2 x^{spy}	0.000 (0.1)	-0.021 (-0.4)	0.040 (1.1)	0.003 (0.9)	-5.321*** (-3.6)
d^v	.	.	.	.	-525.690*** (-32.2)
L1 d^v	0.028 (1.3)	1.665** (2.1)	-0.176 (-0.4)	0.021 (0.4)	16.631 (0.8)
L2 d^v	0.012 (0.5)	1.566 (1.4)	0.273 (0.5)	0.031 (0.3)	21.031 (0.9)
d^u	.	.	.	.	.
L1 d^u	0.000 (0.6)	0.007*** (5.7)	-0.002** (-2.1)	0.000 (0.2)	0.051 (1.5)
L2 d^u	0.000 (0.1)	0.003** (2.2)	-0.001 (-0.7)	-0.000 (-0.1)	0.042 (1.1)
Constant	0.088*** (15.4)	1.522*** (3.8)	-0.256* (-1.7)	-0.043** (-2.1)	3.287 (0.5)
Observations	1,494	1,494	1,494	1,494	1,494
R-squared	0.051	0.092	0.055	0.250	0.766

Table 5 Explanatory power of price and volatility of the S&P500 on option price changes at the hourly frequency.

This table shows the extent to which option price changes can be explained by the two factors as proposed by regression (6). The dependent variable is the change in an option's quoted midpoint, which is regressed on the particular options delta times the changes in the underlying (Δd^s) and the options vega times the change in the volatility in (νd^v). The midpoint price and the change in the underlying and the volatility are sampled at the hourly frequency (the dataset is balanced). We add two lags to be consistent with the previous analyses. The sample uses the full sample of option data of 2014, including options with all expiration dates, strike prices and puts and calls. Column (1) shows results for the full sample, and columns (2)-(10) for various subsets of options sorted by time-to-maturity (TTM) and moneyness (see Table (2)). The first rows show the average TTM and Moneyness of the particular sample. Inference is based on Newey-West standard errors with two lags. The superscripts ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Full Sample	Subsets								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Avg TTM (years)	0.18	0.06	0.05	0.05	0.17	0.16	0.16	0.33	0.32	0.34
Avg Moneyness (S/K)	1.16	0.91	0.98	1.30	0.86	0.98	1.31	0.82	0.98	1.49
Δd^s	1.000*** (5,323.3)	0.622*** (24.7)	0.993*** (303.7)	0.999*** (3,768.2)	1.084*** (163.3)	0.996*** (422.3)	0.999*** (3,812.8)	1.059*** (152.3)	0.987*** (271.8)	1.002*** (2,226.8)
L1 Δd^s	-0.003*** (-23.6)	-0.046*** (-3.1)	0.006*** (3.2)	-0.002*** (-10.0)	0.021*** (4.6)	0.015*** (11.1)	-0.004*** (-22.2)	0.030*** (7.4)	0.021*** (10.2)	-0.006*** (-16.6)
L2 Δd^s	-0.000 (-1.3)	0.036** (2.4)	-0.001 (-0.4)	0.001*** (6.3)	0.005 (1.4)	0.001 (1.2)	-0.001*** (-5.0)	-0.002 (-0.6)	-0.003* (-1.8)	0.000 (0.8)
νd^v	0.902*** (324.6)	1.241*** (56.1)	0.917*** (155.5)	0.914*** (149.9)	0.875*** (131.5)	0.861*** (154.4)	0.871*** (177.8)	0.977*** (88.6)	0.942*** (86.5)	0.969*** (110.5)
L1 νd^v	-0.029*** (-18.2)	0.090*** (7.1)	-0.013*** (-3.5)	-0.015*** (-3.5)	-0.023*** (-4.7)	-0.026*** (-7.1)	-0.041*** (-13.3)	-0.010 (-1.4)	-0.015** (-2.2)	-0.032*** (-5.8)
L2 νd^v	0.026*** (16.3)	0.017 (1.5)	-0.001 (-0.3)	0.003 (1.0)	0.044*** (11.3)	0.020*** (6.1)	0.028*** (9.5)	0.043*** (7.2)	0.027*** (4.1)	0.036*** (6.8)
Constant	0.012*** (50.0)	0.008*** (13.0)	0.006*** (5.5)	0.004*** (7.5)	0.013*** (40.9)	0.018*** (18.8)	0.013*** (27.3)	0.014*** (24.4)	0.019*** (9.9)	0.013*** (16.8)
Observations	851,051	17,579	25,235	122,453	135,154	70,347	256,169	79,283	25,775	119,056
R-squared	0.992	0.863	0.963	0.995	0.935	0.959	0.994	0.917	0.955	0.993

Table 6 Option tick sizes versus standard deviation of price changes.

For each option we calculate the tick size, the average dollar bid-ask spread, and the standard deviation of the price changes (in dollars) sampled at the one-hour, half-hour and one-minute frequency. We then divide options into eight price categories, and we report the median of each variable across the options in each category.

	Option price category								Full Sample
	0 – 1	1 – 5	5 – 10	10 – 20	20 – 40	40 – 100	100 – 500	500 – ∞	
Median Tick size (\$)	0.05	0.05	0.05	0.05	0.05	0.10	0.10	0.10	0.10
Median Spread (\$)	0.35	0.70	1.00	1.27	1.56	2.30	2.94	2.77	2.31
Number of Options	257	951	397	382	433	676	1823	935	5854
Average standard deviation of dollar price changes at different sampling frequencies									
Hour	0.14	0.19	0.46	0.83	1.45	2.65	4.66	7.75	3.23
Half-hour	0.10	0.13	0.32	0.59	1.02	1.9	3.45	5.75	2.37
Minute	0.05	0.03	0.06	0.1	0.17	0.29	0.44	0.49	0.2

Table 7 Explanatory power of price and volatility of the S&P500 on option price changes at the one-minute frequency.

This table is identical to Table 5, but uses data sampled at the one-minute frequency. We regress option price changes on the particular option's delta times the changes in the underlying (Δd^s) and the options vega times the change in the volatility (νd^v). We add ten lags of these variables. Due to the sheer size, we only use data of the first two months of 2014 rather than the whole year (results are unaffected). Column (1) shows results for the full sample, and columns (2)-(10) for various subsets of options sorted by time-to-maturity (TTM) and moneyness. The first rows show the average TTM and Moneyness of the particular sample. Inference is based on Newey-West standard errors with two lags. The superscripts ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Full Sample	Subsets								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Avg TTM (years)	0.1769	0.0629	0.0532	0.0479	0.1672	0.1617	0.159	0.3343	0.3239	0.3572
Avg Moneyness (S/K)	1.1156	0.8818	0.9764	1.305	0.8508	0.9749	1.2688	0.8133	0.9747	1.7148
Δd^s	0.632*** (910.0)	0.245*** (9.5)	0.525*** (144.9)	0.676*** (570.3)	0.210*** (40.0)	0.451*** (190.0)	0.614*** (523.8)	0.240*** (33.4)	0.427*** (83.1)	0.654*** (333.2)
L1 Δd^s	0.247*** (478.5)	0.081*** (4.6)	0.329*** (115.7)	0.247*** (267.1)	0.083*** (26.5)	0.311*** (199.0)	0.241*** (303.4)	0.126*** (24.9)	0.305*** (95.3)	0.248*** (177.8)
L2 Δd^s	0.031*** (69.4)	0.015 (1.3)	0.090*** (40.6)	0.022*** (27.4)	0.016*** (7.0)	0.103*** (86.7)	0.030*** (46.9)	0.033*** (9.5)	0.107*** (44.4)	0.033*** (27.1)
L3 Δd^s	0.017*** (40.8)	-0.003 (-0.2)	0.043*** (19.8)	0.014*** (18.2)	0.010*** (4.3)	0.057*** (48.3)	0.016*** (26.7)	0.024*** (7.6)	0.066*** (30.7)	0.016*** (13.9)
L4-L10 (not shown for brevity) νd^v	0.492*** (13.6)	1.379*** (48.7)	0.707*** (55.4)	0.069 (1.3)	1.475*** (96.5)	0.640*** (20.2)	0.054 (0.6)	1.341*** (28.5)	0.812*** (7.5)	0.612*** (3.0)
L1 νd^v	0.203*** (27.7)	-0.076*** (-4.5)	0.150*** (14.4)	0.345*** (17.4)	0.004 (0.6)	0.208*** (24.4)	0.288*** (17.7)	0.094*** (6.2)	0.222*** (9.3)	0.268*** (7.6)
L2 νd^v	0.007* (1.8)	-0.035*** (-2.9)	0.014* (1.7)	0.013 (0.9)	-0.063*** (-9.8)	0.056*** (11.4)	0.019** (2.3)	-0.018 (-1.5)	0.063*** (5.1)	-0.040** (-2.3)
L3 νd^v	0.004 (1.3)	-0.018 (-1.2)	0.001 (0.2)	0.001 (0.1)	-0.025*** (-4.4)	0.015*** (3.4)	0.015** (2.0)	-0.012 (-1.2)	0.038*** (3.6)	0.001 (0.0)
L4 νd^v	0.004 (1.4)	-0.026* (-1.8)	-0.010 (-1.4)	0.020 (1.5)	-0.024*** (-4.6)	0.020*** (4.8)	-0.006 (-0.9)	0.010 (1.0)	0.030*** (3.1)	0.013 (1.0)
L4-L10 (not shown for brevity)										
Observations	10,169,421	393,753	363,413	1,314,021	2,342,713	1,027,529	2,653,035	910,873	229,741	934,343
R-squared	0.479	0.421	0.506	0.562	0.288	0.436	0.481	0.191	0.384	0.409

Figure 1 Impulse response functions VAR: the impact of order flows on the underlying and volatility returns

This figure plots the IRFs of order flow shocks to the difference of the underlying (d^u) and volatility (d^v). The d^u is based on the SPY ETF quoted midpoint (in cents) and the d^v is based on the VIX extracted from the cross section of option prices (in percentage points). The net order flow (buyer initiated minus seller initiated) in each option is decomposed into dollar order flow exposure to the underlying (x^Δ) based on its delta, and volatility (x^v) based on its vega. The order flows are aggregated across options with all strike prices, expiration dates, and puts and calls. The order flow in the SPY ETF is also added in a separate equation. The IRFs are based on the VAR system of Equation (4) and corresponds to the results in Table 4.

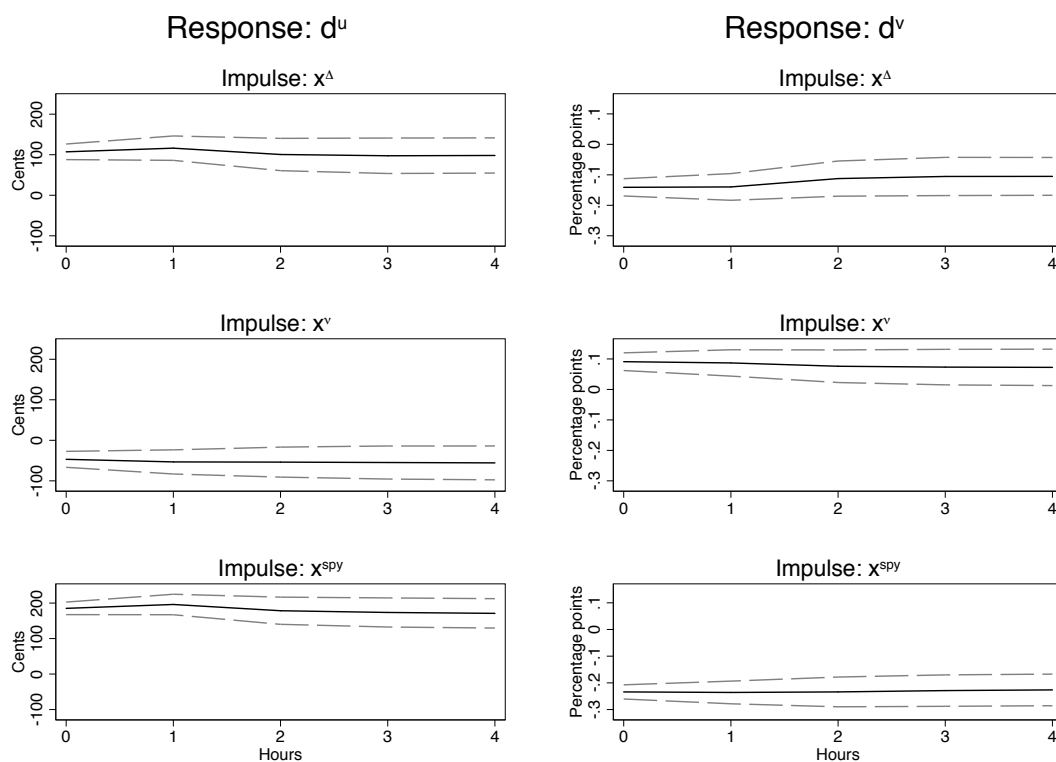


Figure 2 Impulse response functions VAR: half-hour frequency

This figure is identical to Figure 1, but shows results for data sampled at the half-hour frequency. For comparison to the main specification (at the hourly level), we add four lags in the VAR system.

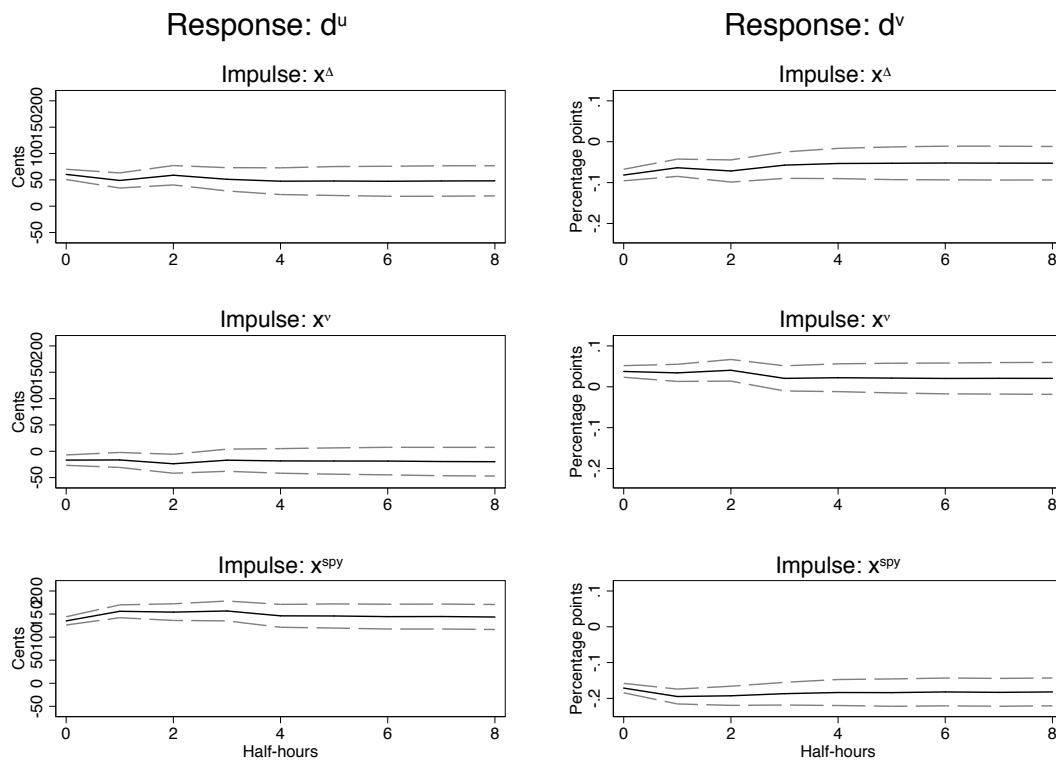
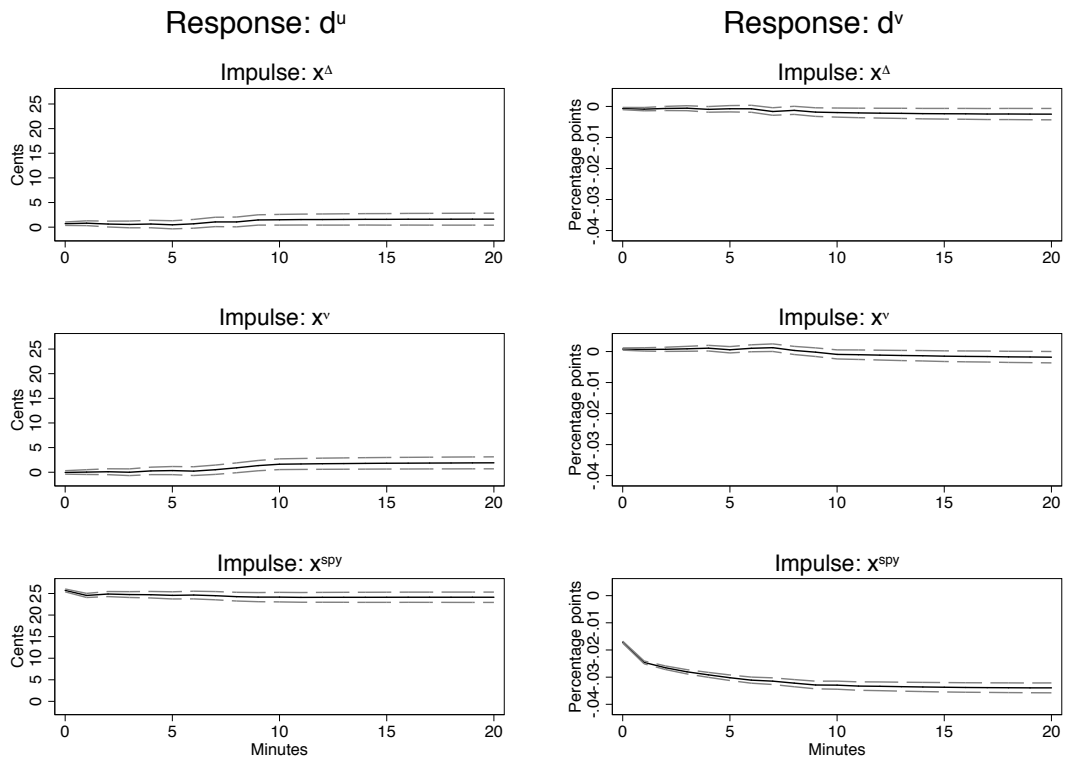


Figure 3 Impulse response functions VAR: one-minute frequency

This figure is identical to Figure 1, but shows results for data sampled at the one-minute frequency. For comparison to the main specification (at the hourly level), we add ten lags in the VAR system and iterate the IRFs for 20 steps (minutes).



Web Appendix to
“Informed Trading in the Index Option Market”

Figure A1 Impulse response functions VAR: Different ordering (1)

This figure plots the IRFs of the VAR model. It is similar to Figure (1) in the paper, except that the ordering of the equations is changed. It follows: $\{x^\nu, x^\Delta, x^{spy}, d^u, d^v\}$.

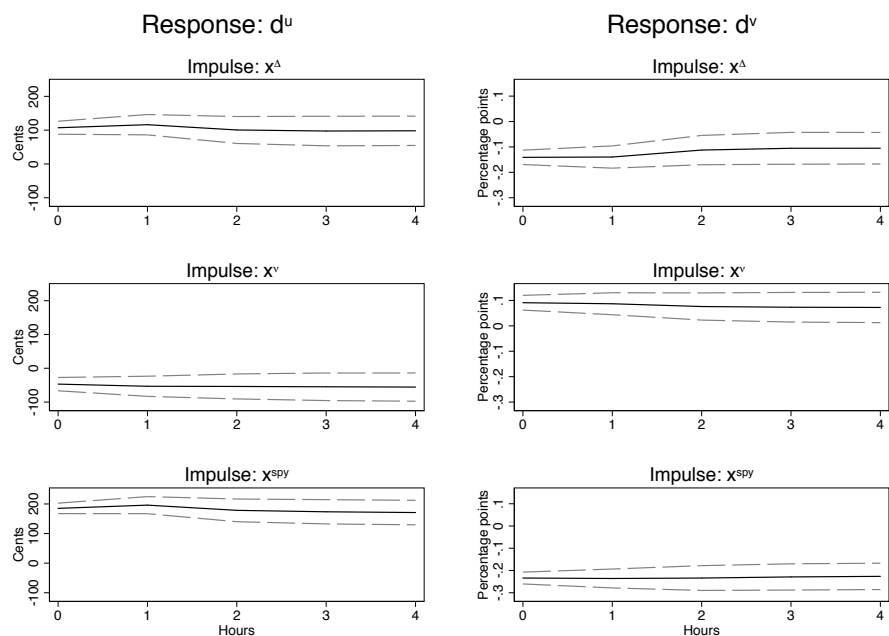


Figure A2 Impulse response functions VAR: Different ordering (2)

This figure plots the IRFs of the VAR model. It is similar to Figure (1) in the paper, except that the ordering of the equations is changed. It follows: $\{x^{spy}, x^\nu, x^\Delta, d^v, d^u\}$.

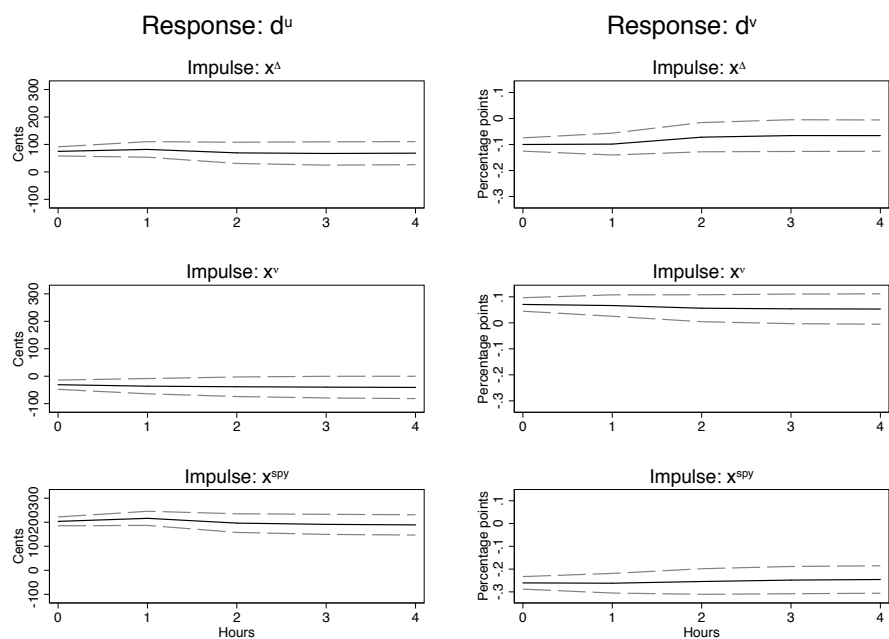


Figure A3 Impulse response functions VAR: forward price instead of SPY ETF price

This figure plots the IRFs of VAR model where we use the price change of the forward (extracted from the cross section of option prices), instead of the SPY ETF price. Otherwise, the figure is identical to Figure (1) in the paper.

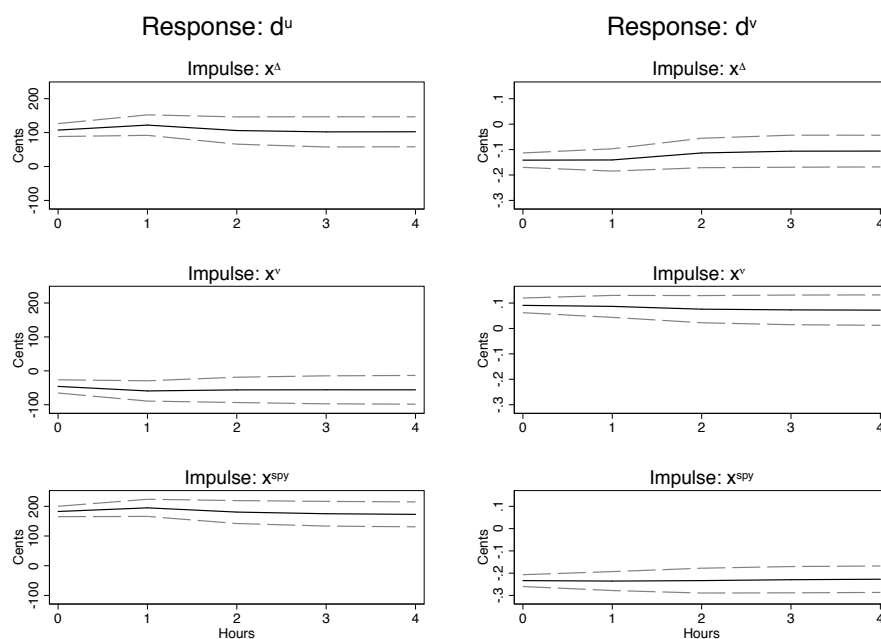


Figure A4 Impulse response functions VAR: returns instead of price changes

This figure plots the IRFs of the VAR model where we use the *return* of the underlying (r^u) and volatility (r^v), instead of the differences. Otherwise, the figure is identical to Figure (1) in the paper.

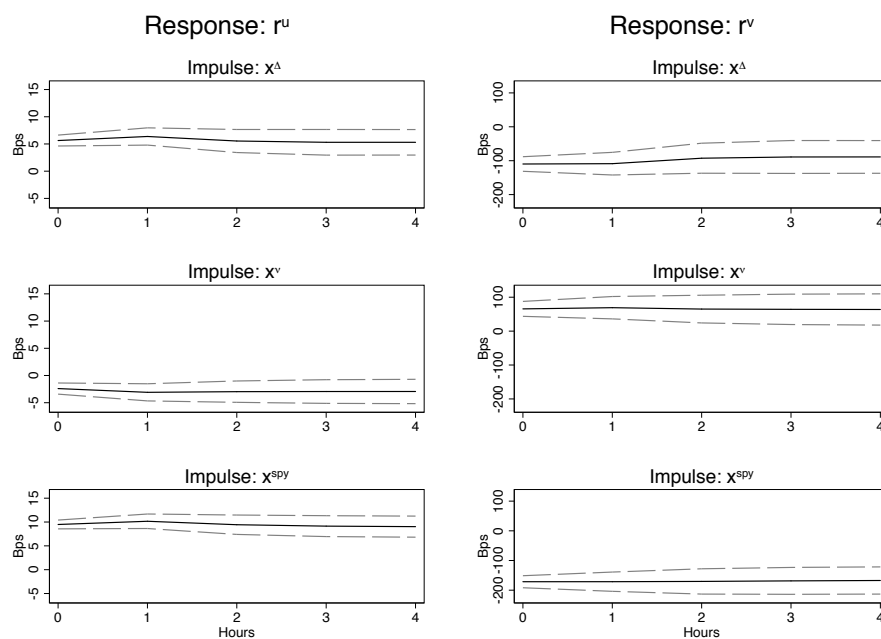


Figure A5 Impulse response functions VAR: No SPY ETF order flow

This figure plots the IRFs of a four-equation VAR model, where we omit the spy ETF order flow equation altogether.

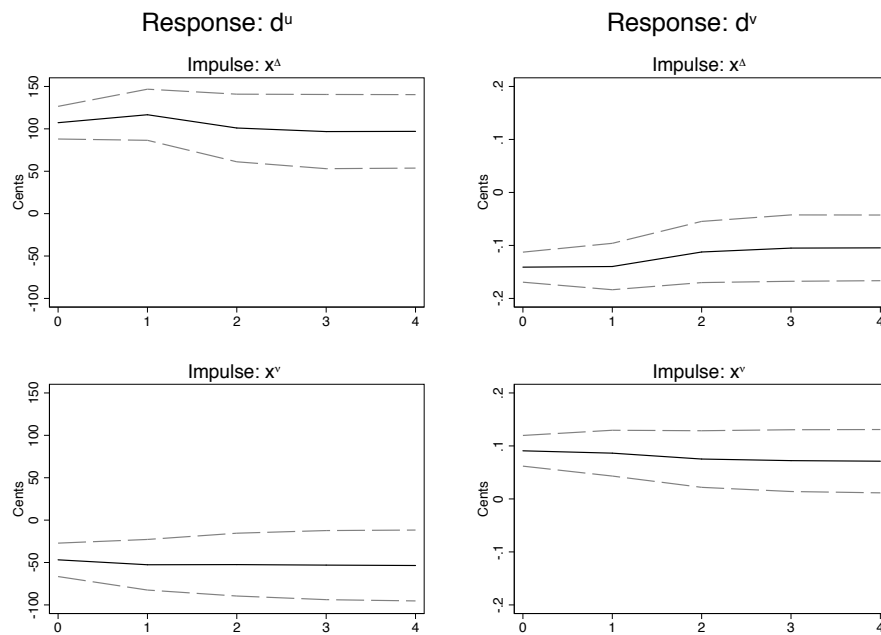


Table 1 Explanatory power of S&P500 underlying and volatility on option price changes at the half-hour frequency.

This table is identical to Table 5, but uses data sampled at the half-hour frequency. It shows the extent to which option price changes can be explained by the two underlying factors as proposed in Equation (6). The dependent variable is the change in an option price, which is regressed on the particular options delta times the changes in the underlying (Δd^s), and the options vega times the change in the volatility (νd^v). We add four lags to be consistent with the previous analyses. Column (1) shows results for the full sample, and columns (2)-(10) for various subsets of options sorted by time-to-maturity (TTM) and moneyness. The first rows show the average TTM and Moneyness of the particular sample. Inference is based on Newey-West standard errors with two lags. The superscripts ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Avg TTM (years)	0.18	0.06	0.05	0.05	0.17	0.16	0.16	0.33	0.32	0.33
Avg Moneyness (S/K)	1.25	0.90	0.98	1.39	0.85	0.98	1.39	0.82	0.98	1.69
Δd^s	0.986*** (3.965.5)	0.312*** (3.6)	0.956*** (392.7)	0.983*** (2,658.4)	0.477*** (19.9)	0.953*** (537.5)	0.990*** (3,054.3)	0.617*** (27.9)	0.939*** (355.1)	0.990*** (2,211.2)
L1 Δd^s	0.019*** (88.1)	-0.013 (-0.3)	0.052*** (23.6)	0.021*** (58.9)	0.022* (2.0)	0.074*** (47.1)	0.017*** (53.6)	0.047*** (4.9)	0.081*** (35.4)	0.018*** (39.2)
L2 Δd^s	0.004*** (20.3)	0.060** (2.3)	0.001 (0.7)	-0.003*** (-8.8)	0.008 (0.9)	0.008*** (5.7)	0.006*** (21.7)	0.008 (0.9)	0.011*** (5.3)	0.005*** (13.4)
L3 Δd^s	-0.002*** (-10.6)	-0.006 (-0.2)	-0.000 (-0.1)	0.002*** (7.5)	-0.010 (-1.1)	-0.008*** (-6.8)	-0.003*** (-11.4)	-0.006 (-0.7)	-0.006*** (-3.0)	-0.002*** (-5.9)
L4 Δd^s	0.001*** (4.1)	0.045* (1.8)	-0.002 (-1.3)	0.003*** (10.9)	-0.001 (-0.1)	-0.002 (-1.3)	-0.000* (-1.9)	0.004 (0.4)	-0.005*** (-2.8)	0.001*** (3.5)
νd^v	0.896*** (474.3)	1.422*** (25.8)	0.927*** (179.3)	0.870*** (120.2)	1.109*** (83.3)	0.859*** (216.9)	0.848*** (241.0)	1.051*** (90.1)	0.926*** (124.4)	0.966*** (167.1)
L1 νd^v	0.001 (0.9)	-0.006 (-0.2)	-0.017*** (-3.9)	0.004 (0.6)	-0.005 (-0.7)	0.023*** (7.7)	-0.011*** (-3.3)	0.028*** (3.8)	0.051*** (8.9)	-0.003 (-0.6)
L2 νd^v	-0.025*** (-16.4)	0.009 (0.6)	-0.015*** (-3.8)	-0.002 (-0.4)	-0.021*** (-2.8)	-0.027*** (-8.1)	-0.019*** (-6.7)	-0.035*** (-5.0)	-0.029*** (-5.4)	-0.030*** (-7.3)
L3 νd^v	-0.010*** (-7.9)	0.029 (1.5)	0.000 (0.1)	-0.007 (-1.2)	-0.007 (-1.3)	-0.015*** (-5.9)	-0.009*** (-3.4)	-0.016*** (-2.8)	-0.023*** (-5.2)	-0.005 (-1.6)
L4 νd^v	0.006*** (4.4)	-0.024 (-1.4)	-0.006** (-2.0)	0.007 (1.3)	-0.001 (-0.1)	0.003 (1.0)	0.008*** (3.1)	0.009 (1.5)	0.005 (1.2)	0.003 (0.8)
Constant	0.004*** (18.8)	0.002*** (6.7)	0.001 (1.1)	0.003*** (5.5)	0.002*** (6.1)	0.008*** (12.1)	0.004*** (9.6)	0.002*** (4.2)	0.008*** (6.4)	0.003*** (5.4)
Observations	2,165,720	49,481	53,443	317,013	312,690	143,162	720,471	180,229	55,436	333,795
R-squared	0.968	0.796	0.921	0.973	0.757	0.906	0.971	0.683	0.895	0.972